BITS Pilani

Pilani|Dubai|Goa|Hyderabad

CE F231
Fluid Mechanics

Sayantan Chakraborty, Ph.D., AM ASCE, MIE
sayantan.chakraborty@pilani.bits-pilani.ac.in

innovate achieve lead

BITS Pilani

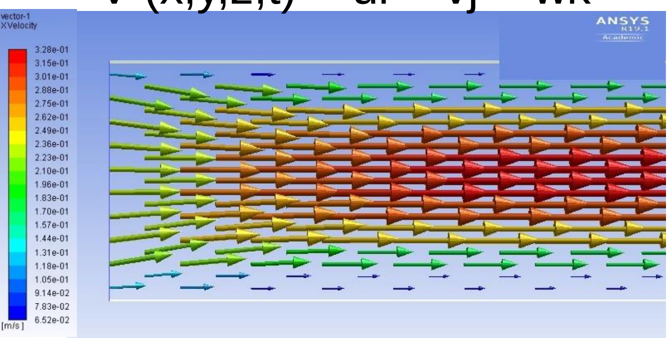
Pilani|Dubai|Goa|Hyderabad

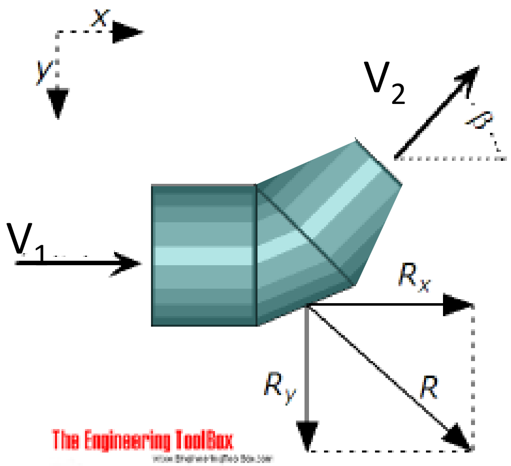
Lecture 17: Fluid Kinematics

Continuity Equation
(Mathematical statement/equation for conservation of mass)

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0}$$

V (x,y,z,t) = ui + vj + wk





https://www.engineeringtoolbox.com/forces-pipe-bends-d_968.html

Reducing elbow

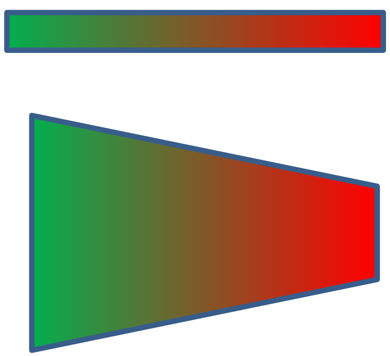
https://www.zoro.com/zoro-select-reducing-elbow-90-deg-10x8-125-flg-blk-4319001766/i/G1557031/

Above: Typical example of thrust block in situ.

https://www.utilitystrikeavoidancegroup.org/safety-alerts.html

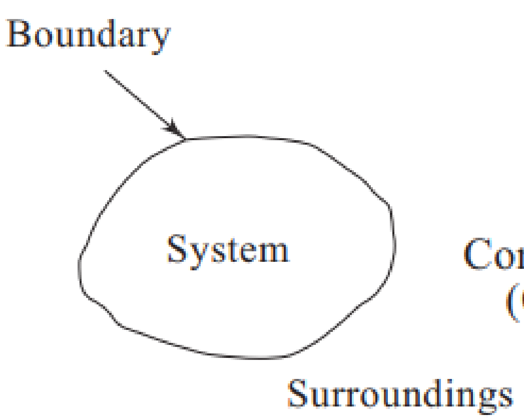
http://www.mepskills.com/2020/10/how-to-use-venturi-meter-for-measuring.html

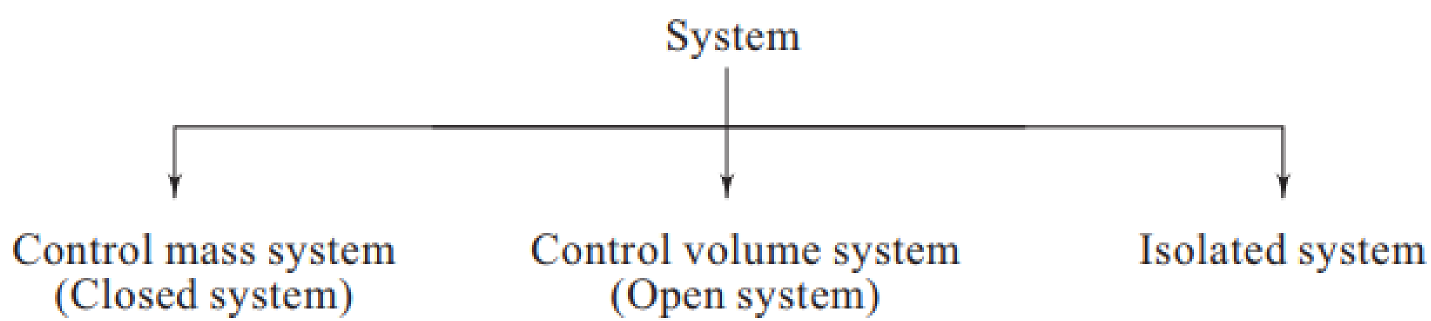
Low vel. High vel.



System

- System
  - A system is defined as a quantity of matter in space on which attention is paid in the analysis of problem
  - Everything outside system is called surroundings

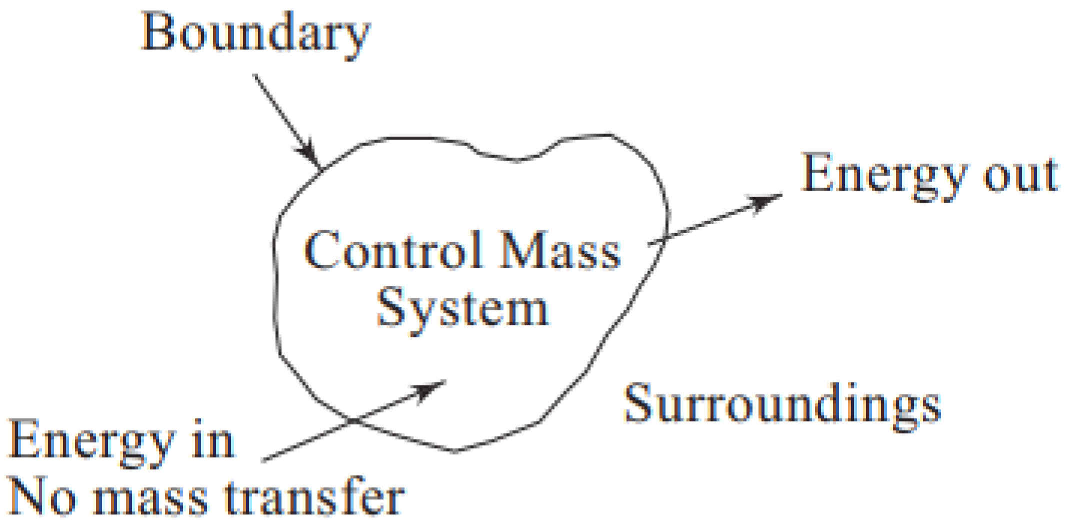




Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

Control Mass

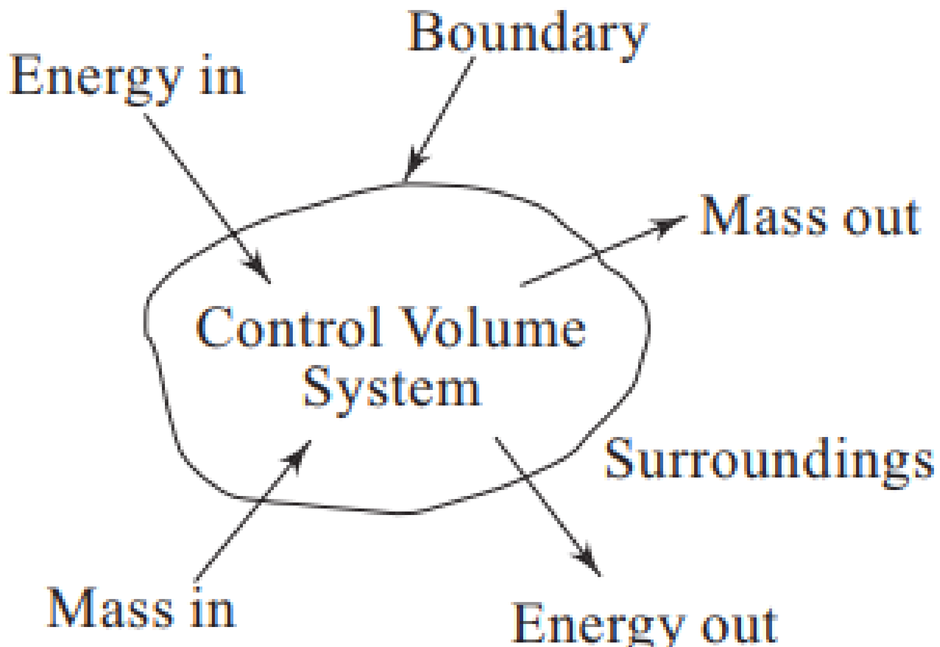
- Control Mass System
  - System with fixed mass
  - Mass transfer not allowed through boundary
  - Volume may change – boundary may shrink or expand
  - Control mass system is aka Closed system



Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

Control Volume

- Control Volume System
  - System with fixed volume
  - Mass transfer allowed through boundary
  - Control volume system is aka Open system

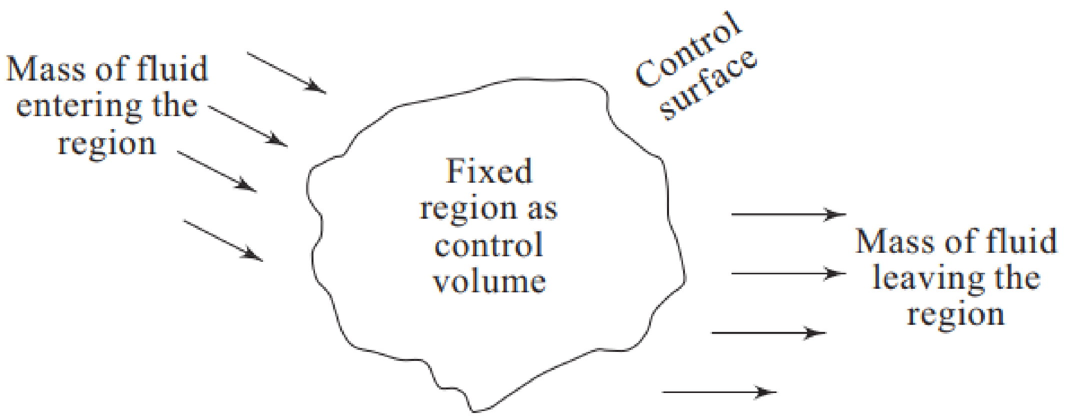


Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

Conservation of Mass – The Continuity Equation

- Control mass system – conservation of mass is inherently satisfied

- Control volume system

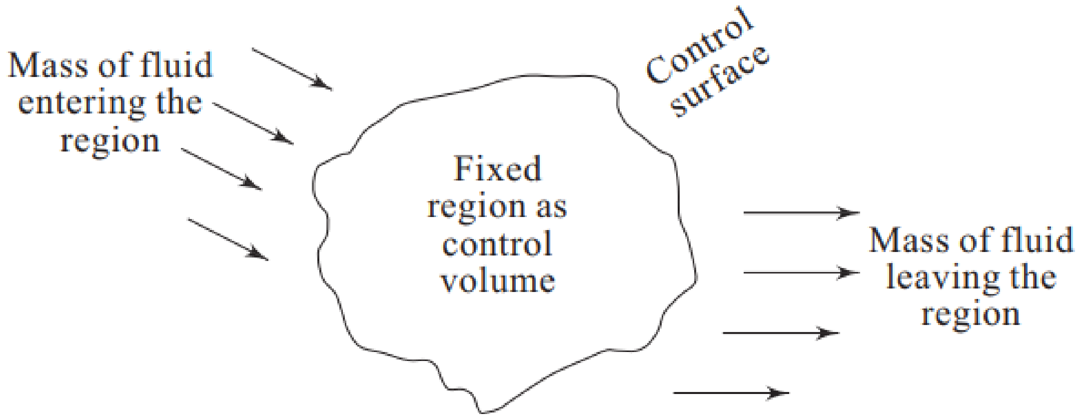


Rate at which mass enters the region	= Rate at which mass leaves the region + Rate of accumulation of mass in the region

Salary (Rs. 1,00,000/month) = Expenditure (Rs. 60,000) + Savings (Rs. 40,000)

Conservation of Mass – The Continuity Equation

- Control volume system



Rate at which mass enters the region = Rate at which mass leaves the region + Rate of accumulation of mass in the region

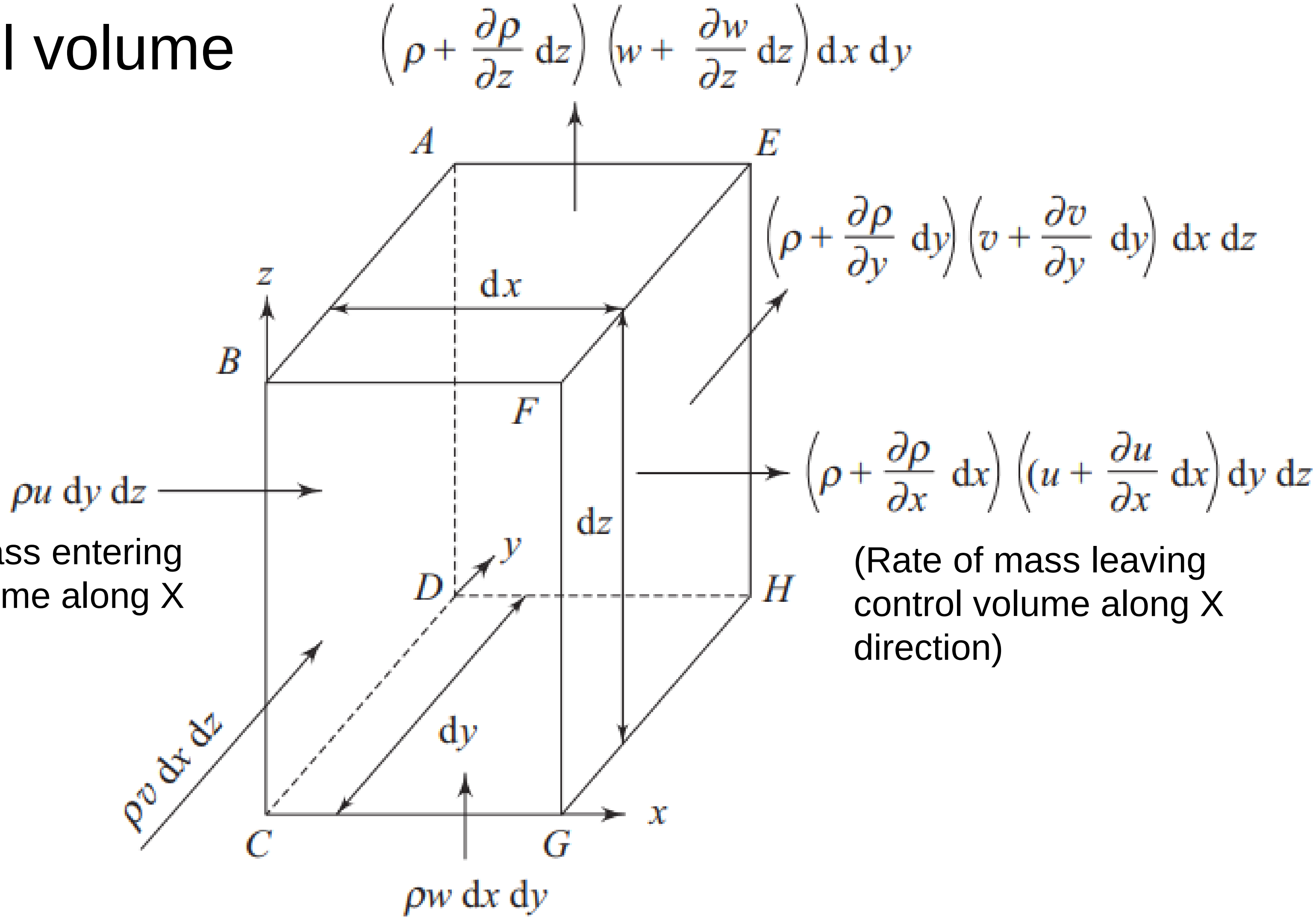
$$\boxed{\text{Rate of accumulation of mass in the control volume} + \text{Net rate of mass efflux from the control volume} = 0}$$

Savings (Rs. 40,000) + Expenditure (Rs. 60,000) - Salary (Rs. 1,00,000) = 0

Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

Conservation of Mass – The Continuity Equation

- Control volume



(Rate of mass entering control volume along X direction)

(Rate of mass leaving control volume along X direction)

Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

Conservation of Mass – The Continuity Equation

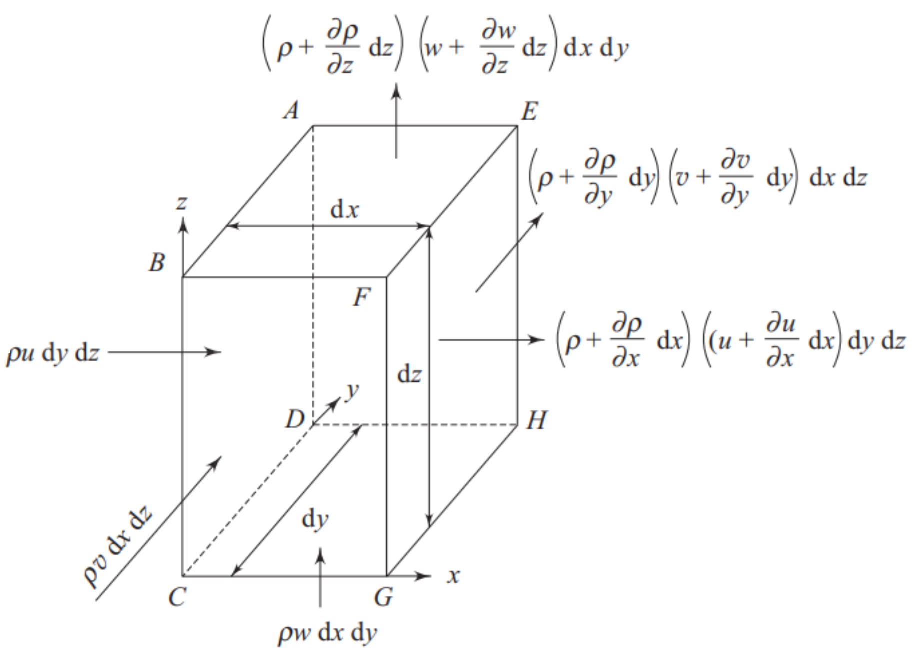
- Rate of mass entering through ABCD = $\rho u \, \mathrm{d}y \, \mathrm{d}z$

- Rate of mass leaving EFGH

$$= \left(\rho + \frac{\partial \rho}{\partial x}\mathrm{d}x\right)\left(u + \frac{\partial u}{\partial x}\mathrm{d}x\right)\mathrm{d}y\,\mathrm{d}z$$

$$= \left[\rho u + \frac{\partial}{\partial x}(\rho u)\,\mathrm{d}x\right]\mathrm{d}y\,\mathrm{d}z$$

(neglecting the higher order terms in $\mathrm{d}x$)



Conservation of Mass – The Continuity Equation

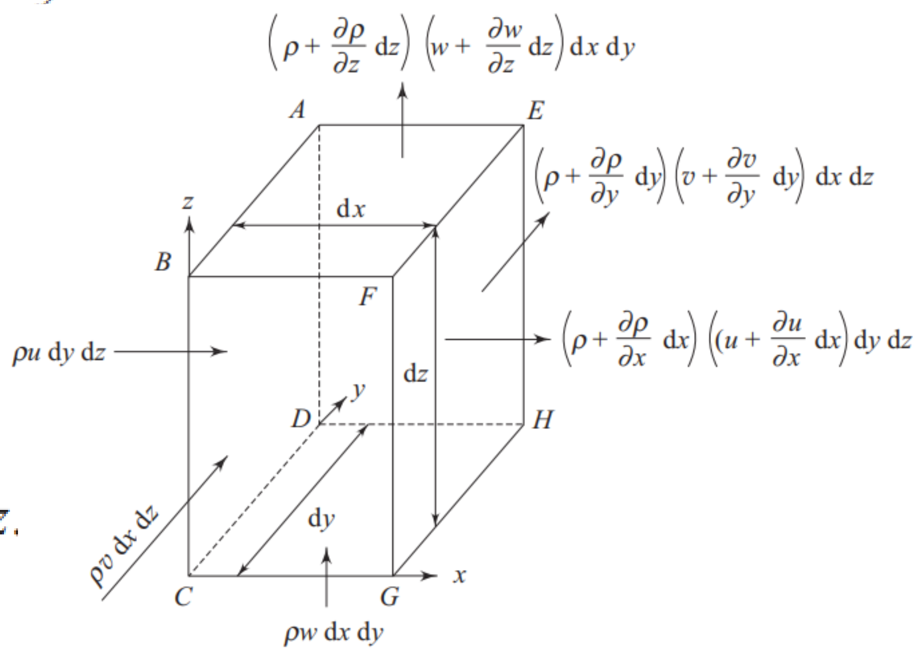
- Net rate of mass efflux from control volume in x-direction

$$= \left[\rho u + \frac{\partial}{\partial x}(\rho u)\,\mathrm{d}x\right]\mathrm{d}y\,\mathrm{d}z - \rho u\,\mathrm{d}y\,\mathrm{d}z$$

$$= \frac{\partial}{\partial x}(\rho u)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$$

$$= \frac{\partial}{\partial x}(\rho u)\,\mathrm{d}\forall$$

where $\mathrm{d}\forall$ is the elemental volume $\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$.



Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

Conservation of Mass – The Continuity Equation

In a similar fashion, the net rate of mass efflux in the y direction

$$= \left(\rho + \frac{\partial \rho}{\partial y}\,\mathrm{d}y\right)\left(v + \frac{\partial v}{\partial y}\,\mathrm{d}y\right)\mathrm{d}x\,\mathrm{d}z - \rho v\,\mathrm{d}x\,\mathrm{d}z$$

$$= \frac{\partial}{\partial y}(\rho v)\,\mathrm{d}\mathcal{V}$$

and, the net rate of mass efflux in the z direction

$$= \left(\rho + \frac{\partial \rho}{\partial z}\,\partial z\right)\left(w + \frac{\partial w}{\partial z}\,\partial z\right)\mathrm{d}x\,\mathrm{d}y - \rho w\,\mathrm{d}x\,\mathrm{d}y$$

$$= \frac{\partial}{\partial z}(\rho w)\,\mathrm{d}\mathcal{V}$$

Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

Conservation of Mass – The Continuity Equation

innovate achieve lead

$$\boxed{\text{Rate of accumulation of mass in the control volume + Net rate of mass efflux from the control volume} = 0}$$

The rate of accumulation of mass within the control volume is $\frac{\partial}{\partial t}(\rho \, d\mathcal{V}) = \frac{\partial \rho}{\partial t} d\mathcal{V}$ (by the definition of control volume, $d\mathcal{V}$ is invariant with time).

Along X - direction

$$\frac{\partial}{\partial x}(\rho u)\, d\mathcal{V}$$

Along Y - direction

$$\frac{\partial}{\partial y}(\rho v)\, d\mathcal{V}$$

Along Z - direction

$$\frac{\partial}{\partial z}(\rho w)\, d\mathcal{V}$$

$$\left\{\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w)\right\} d\mathcal{V} = 0$$

Conservation of Mass – The Continuity Equation

- Continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0 \quad \Rightarrow \quad \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0$$

Applicable for:
1) Steady and unsteady flow
2) Uniform and non-uniform flow
3) Compressible and incompressible fluid flow

Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

Conservation of Mass – The Continuity Equation

- Continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0$$

- Continuity equation for steady flow

$$\frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0 \Rightarrow \nabla \cdot (\rho \vec{V}) = 0$$

Conservation of Mass – The Continuity Equation

- Continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0$$

- Continuity equation for incompressible flow

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad \Rightarrow \quad \nabla \cdot (\vec{V}) = 0$$

(Divergence)

- Physically means volume of a fluid element remains same in its course of its flow

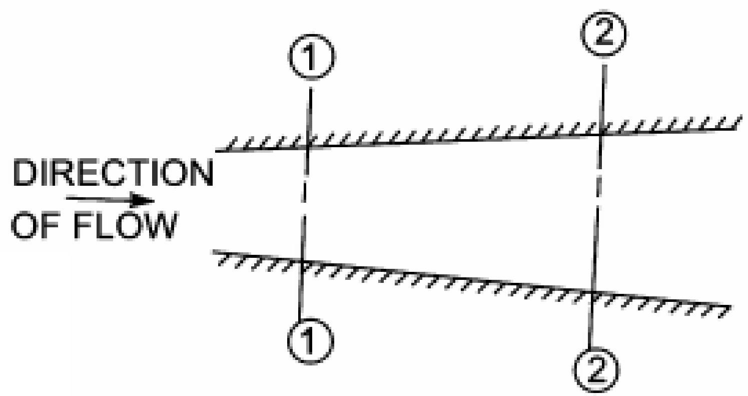
Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

Conservation of Mass – The Continuity Equation

- Continuity equation for 1D incompressible fluid flow

Rate of accumulation of mass in the control volume + Net rate of mass efflux from the control volume = 0

$$\boxed{\frac{\partial}{\partial t}(\rho\, d\forall)}$$



Rate of flow at section 1-1 = $\rho_1 A_1 v_1$

Rate of flow at section 2-2 = $\rho_2 A_2 v_2$

($\rho 1 = \rho 2$ for incompressible flow)

$$\boxed{\text{Rate of flow or Discharge} = A_1 v_1 = A_2 v_2}$$

Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

Continuity Equation: Example

- Continuity equation for 2D incompressible flow

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

In a two-dimensional incompressible flow over a solid plate, the velocity component perpendicular to the plate is $v = 2x^2 y^2 + 3y^3 x$, where x is the coordinate along the plate and y is perpendicular to the plate. Hence find out

(i) the velocity component along the plate

Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

Continuity Equation: Example

- Continuity equation for 3D incompressible flow

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0$$

For the following set of velocity components verify whether the continuity equation is satisfied. If so, determine the (i) vorticity vector and (ii) acceleration vector at point A (1, 1, 1)

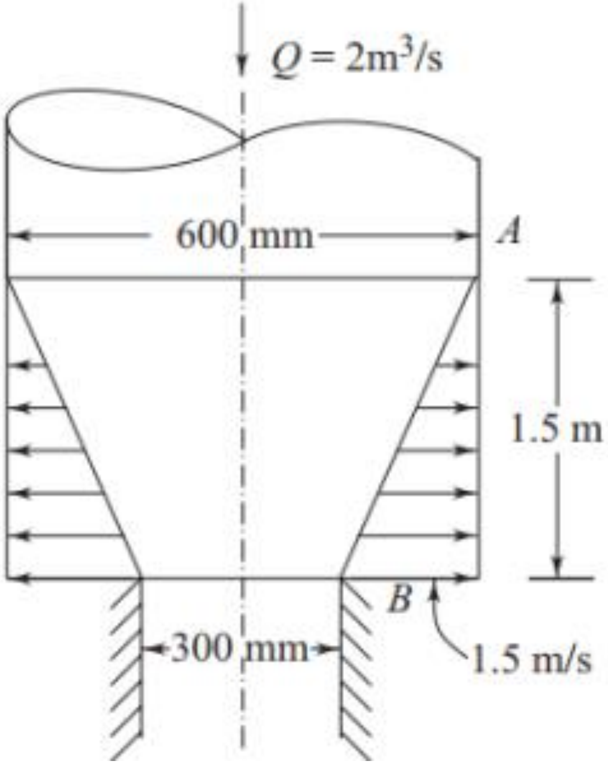
$$u = 2x^2 + 3y$$
$$v = -2xy + 3y^3 + 3zy$$
$$w = -\frac{3}{2}z^2 - 2xz - 9y^2 z$$

Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

Continuity Equation: Example

Water flows downward in a pipe of 600 mm diameter at the rate of 2 m^3/s. It then enters a conical duct (Fig. 4.37) with porous walls such that there is a radial outflow with flow velocity varying linearly from zero at *A* to 1.5 m/s at *B*. What is the rate of flow at *B* coming out from the conical duct.

(*Ans*. 0.587 m^3/s)



Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

Continuity Equation: Example

A conical nozzle 0.5 m long converging from 0.4 m to 0.2 m diameter linearly is subjected to an outlet flow varying linearly from 0 to 1 m^3/s in 10 seconds. Determine the total acceleration at the mid-length of the nozzle at the mid-time of variation. Assume uniform flow over each cross-section.

Find diameter of any section as function of x

Find discharge at outlet as function of t

Find discharge at any section as function of t

Find velocity at any section as function of x and t

Find total acceleration

Som, S.K. and Biswas, C., 2006. Fluid Mechanics: basic Concepts & Principles.

THANK YOU